Airbnb Market Analysis — Barcelona

SQL Project

Total listings
19410



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Data source: insideairbnb.com/get-the-data/

1. Market Overview: The Big Picture

Objective

Understand the size, structure, and distribution of Airbnb listings in Barcelona.

Listings by Neighborhood (Market Concentration)

```
SELECT neighbourhood_cleansed,
 COUNT(*) AS total_listings
 FROM `airbnb-sql-project.airbnb_barcelona.listings`
GROUP BY neighbourhood_cleansed
 ORDER BY total_listings DESC;
```

Job information | Results | Visualisation | JSON

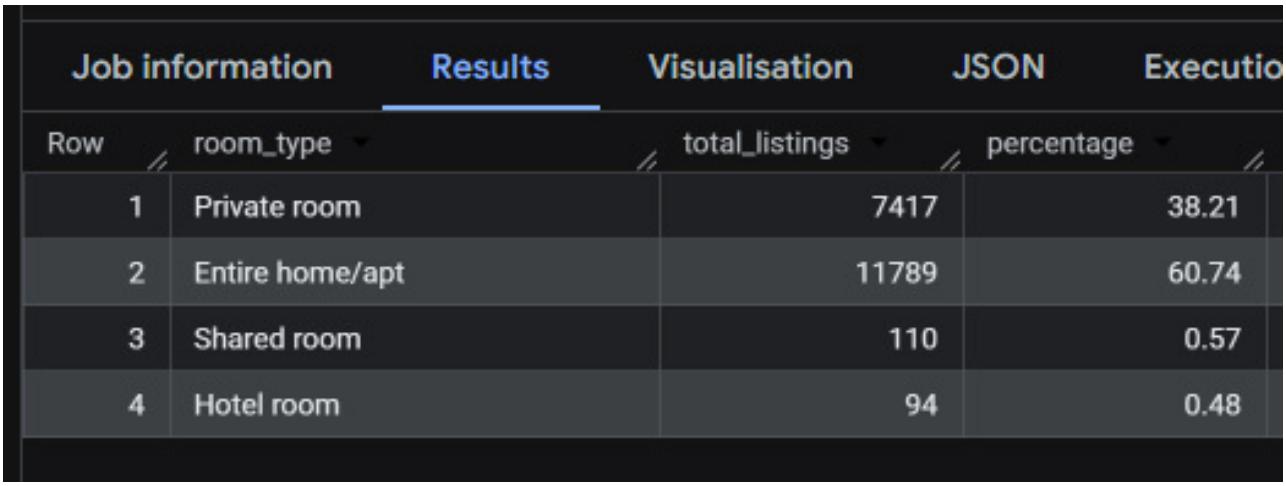
Row	neighbourhood_cleansed	total_listings
1	la Dreta de l'Eixample	2416
2	el Raval	1519
3	el Barri Gòtic	1253
4	Sant Pere, Santa Caterina i la Ri...	1244
5	la Vila de Gràcia	1136
6	la Sagrada Família	1102
7	l'Antiga Esquerra de l'Eixample	1042

Insight: Barcelona's Airbnb market is heavily concentrated in central neighborhoods, creating high competition in tourist areas and lower saturation in outer districts.

1. Market Overview:

Listings by Room Type

```
SELECT room_type,
COUNT(*) AS total_listings,
ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (), 2) AS
percentage
FROM `airbnb-sql-project.airbnb_barcelona.listings`
GROUP BY room_type;
```



Job information | Results | Visualisation | JSON | Executio

Row	room_type	total_listings	percentage
1	Private room	7417	38.21
2	Entire home/apt	11789	60.74
3	Shared room	110	0.57
4	Hotel room	94	0.48

Insight: The Airbnb market in Barcelona is dominated by entire homes/apartments, which account for approximately 61% of all listings, indicating a strong focus on full-property rentals.

2. Pricing Dynamics: The Money In

Price by Neighborhood

```
SELECT neighbourhood_cleansed,
ROUND(AVG(price), 2) AS avg_price,
COUNT(*) AS total_listings
FROM `airbnb-sql-project.airbnb_barcelona.listings`
WHERE price IS NOT NULL
GROUP BY neighbourhood_cleansed
HAVING COUNT(*) >= 100
ORDER BY avg_price DESC;
```



Row	neighbourhood_cleansed	avg_price	total_listings
1	Sants - Badal	328.46	156
2	el Guinardó	265.64	103
3	Diagonal Mar i el Front Marítim ...	261.52	128
4	la Dreta de l'Eixample	252.84	1942
5	el Camp de l'Arpa del Clot	248.8	256
6	la Nova Esquerra de l'Eixample	246.8	639
7	la Vila Olímpica del Poblenou	229.33	134

Insight: Airbnb prices in Barcelona vary across neighborhoods, with the highest averages in areas like Sants - Badal, el Guinardó, and Diagonal Mar.

Central tourist areas such as el Raval and Barceloneta have lower prices, likely due to higher competition.

2. Pricing Dynamics: The Money In

Above-Average Price Neighborhoods

```
SELECT
  neighbourhood_cleansed,
  ROUND(AVG(price), 2) AS avg_price
FROM `airbnb-sql-project.airbnb_barcelona.listings`
WHERE price IS NOT NULL
GROUP BY neighbourhood_cleansed
HAVING AVG(price) > (
  SELECT AVG(price)
  FROM `airbnb-sql-project.airbnb_barcelona.listings`
  WHERE price IS NOT NULL
)
AND COUNT(*) >= 100
ORDER BY avg_price DESC;
```

Row	neighbourhood_cleansed	avg_price
1	Sants - Badal	328.46
2	el Guinardó	265.64
3	Diagonal Mar i el Front Marítim ...	261.52
4	la Dreta de l'Eixample	252.84
5	el Camp de l'Arpa del Clot	248.8
6	la Nova Esquerra de l'Eixample	246.8
7	la Vila Olímpica del Poblenou	229.33

Insight: Several neighborhoods have above-average Airbnb prices, including Sants - Badal, el Guinardó, and Diagonal Mar, along with la Dreta de l'Eixample.

After filtering for sufficient listing volume, these areas represent the most reliable premium segments of the market.

2. Pricing Dynamics: The Money In

Price vs Capacity

```
SELECT
 accommodates,
 ROUND(AVG(price), 2) AS avg_price,
 COUNT(*) AS total_listings
FROM `airbnb-sql-project.airbnb_barcelona.listings`
WHERE price IS NOT NULL
 AND price < (
  SELECT APPROX_QUANTILES(price, 100)[OFFSET(99)]
  FROM `airbnb-sql-project.airbnb_barcelona.listings`
 )
GROUP BY accommodates
ORDER BY accommodates;
```

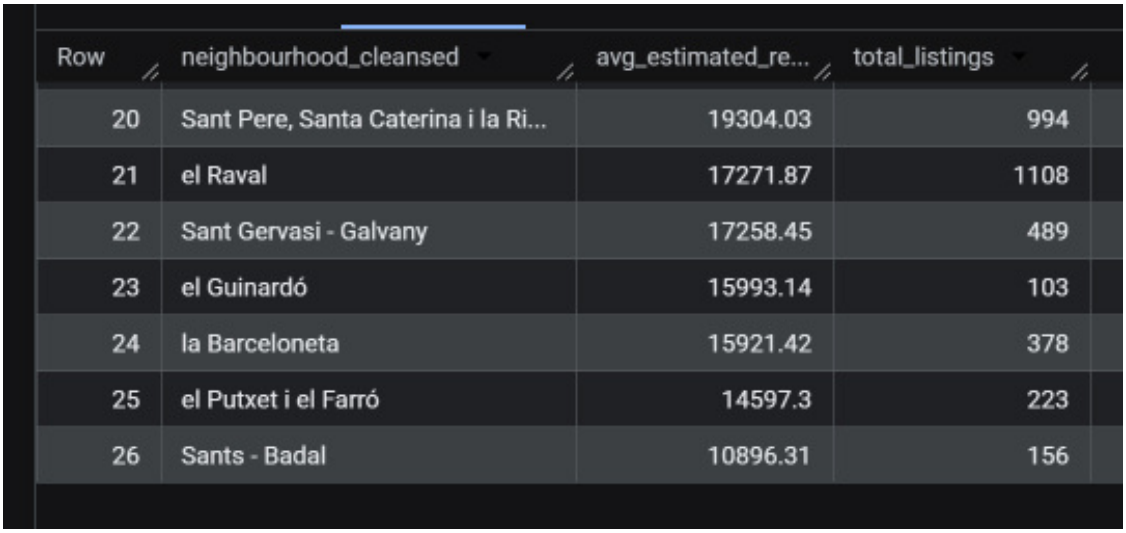
Row	accommodates	avg_price	total_listings
1	1	53.59	2088
2	2	111.59	4253
3	3	148.12	1499
4	4	180.02	3320
5	5	202.9	1344
6	6	235.46	1488
7	7	257.57	312

Insight: Prices rise with capacity, especially up to 8 guests, while larger listings form a smaller, higher-priced luxury segment.

3. Revenue & Market Performance

Estimated Revenue per Listing

```
SELECT
 neighbourhood_cleansed,
 ROUND(AVG(price * (365 - availability_365)), 2) AS
avg_estimated_revenue,
 COUNT(*) AS total_listings
FROM `airbnb-sql-project.airbnb_barcelona.listings`
WHERE price IS NOT NULL
 AND availability_365 IS NOT NULL
GROUP BY neighbourhood_cleansed
HAVING COUNT(*) >= 100
ORDER BY avg_estimated_revenue DESC;
```



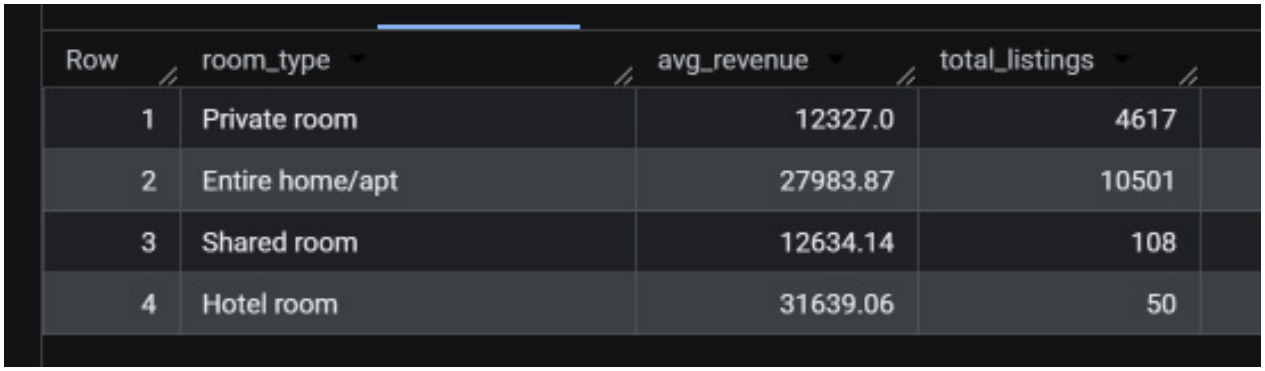
Row	neighbourhood_cleansed	avg_estimated_re...	total_listings
20	Sant Pere, Santa Caterina i la Ri...	19304.03	994
21	el Raval	17271.87	1108
22	Sant Gervasi - Galvany	17258.45	489
23	el Guinardó	15993.14	103
24	la Barceloneta	15921.42	378
25	el Putxet i el Farró	14597.3	223
26	Sants - Badal	10896.31	156

Insight: The most profitable neighborhoods are not necessarily the most central, but those that effectively balance premium pricing with steady booking demand.

3. Revenue & Market Performance

Revenue by Room Type

```
SELECT
  room_type,
  ROUND(AVG(price * (365 - availability_365)), 2) AS
avg_revenue,
  COUNT(*) AS total_listings
FROM `airbnb-sql-project.airbnb_barcelona.listings`
WHERE price IS NOT NULL
  AND availability_365 IS NOT NULL
GROUP BY room_type;
```



Row	room_type	avg_revenue	total_listings
1	Private room	12327.0	4617
2	Entire home/apt	27983.87	10501
3	Shared room	12634.14	108
4	Hotel room	31639.06	50

Insight: 1. Entire homes dominate
- High revenue
- Massive supply

core market segment

2. Hotel rooms = misleading high
- Highest revenue
- Very small sample

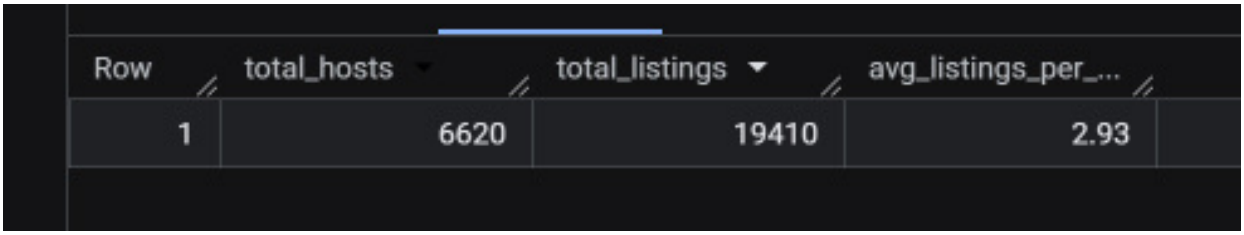
niche segment, not reliable

3. Budget segment
- Private + shared rooms → ~€12k
- lower earning potential

4. Host Behavior: The Competition

Professional Hosts (Multi-listing)

```
SELECT COUNT(DISTINCT host_id) AS total_hosts,
COUNT(*) AS total_listings,
ROUND(COUNT(*) / COUNT(DISTINCT host_id), 2) AS
avg_listings_per_host
FROM airbnb-sql-project.airbnb_barcelona.listings;
```



Row	total_hosts	total_listings	avg_listings_per_...
1	6620	19410	2.93

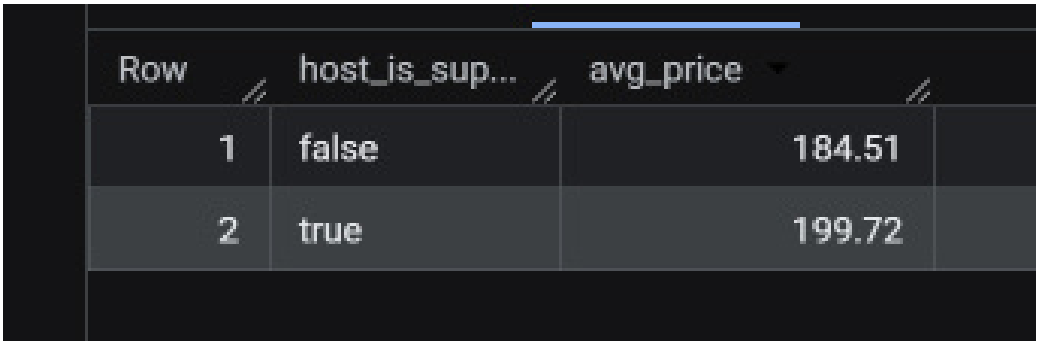
Insight: The Barcelona Airbnb market consists of 6,620 hosts managing 19,410 listings, resulting in an average of 2.93 listings per host.

This indicates that many hosts operate more than one property, suggesting a partially professionalized market rather than one dominated purely by individual, single-property hosts.

5. Superhost Performance

Price Difference

```
SELECT
  host_is_superhost,
  ROUND(AVG(price), 2) AS avg_price
FROM `airbnb-sql-project.airbnb_barcelona.listings`
WHERE price IS NOT NULL
  AND host_is_superhost IS NOT NULL
GROUP BY host_is_superhost;
```



Row	host_is_sup...	avg_price
1	false	184.51
2	true	199.72

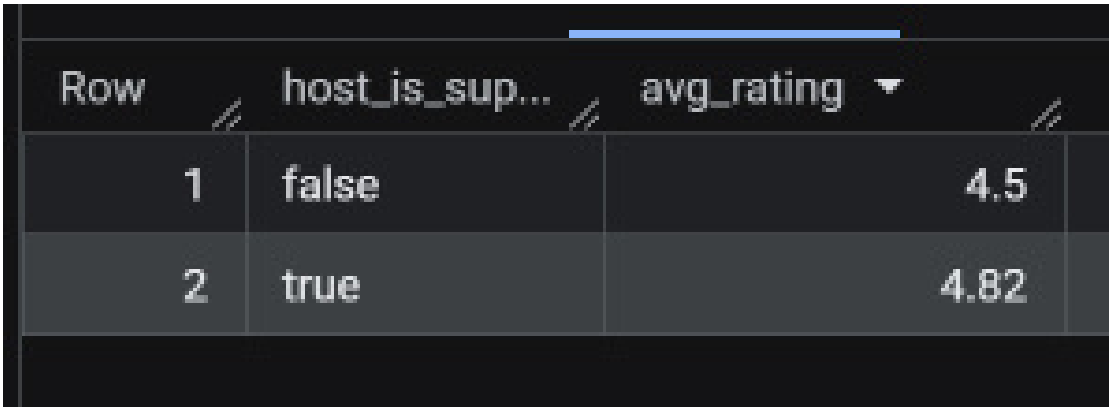
Insight: Superhosts in Barcelona charge higher average prices (€199.72) compared to non-superhosts (€184.51), representing a price premium of approximately €15 per night (~8%).

This suggests that hosts with Superhost status are able to command higher rates, likely due to increased trust, better reviews, and stronger reputations among guests.

5. Superhost Performance

Rating Difference

```
SELECT host_is_superhost, ROUND(AVG(review_scores_rating), 2) AS avg_rating
FROM airbnb-sql-project.airbnb_barcelona.listings
WHERE review_scores_rating IS NOT NULL
 AND host_is_superhost IS NOT NULL
GROUP BY host_is_superhost;
```



Row	host_is_sup...	avg_rating
1	false	4.5
2	true	4.82

Insight: Superhosts in Barcelona achieve significantly higher average ratings (4.82) compared to non-superhosts (4.50), indicating consistently better guest experiences.

This substantial difference highlights the effectiveness of the Superhost designation in identifying high-quality hosts who deliver superior service.

6. Demand & Guest Behavior

Underpriced Listings (High Demand + Low Price)

```
WITH city_avg AS (
  SELECT AVG(price) AS avg_price
  FROM `airbnb-sql-project.airbnb_barcelona.listings`
  WHERE price IS NOT NULL
)

SELECT
  name,
  neighbourhood_cleansed,
  price,
  availability_365,
  number_of_reviews
FROM `airbnb-sql-project.airbnb_barcelona.listings`, city_avg
WHERE price < avg_price
  AND availability_365 BETWEEN 1 AND 100
  AND number_of_reviews > 50
ORDER BY availability_365 ASC;
```



Row	name	neighbourhood_cleansed	price	availability_365	number_of_reviews
1	Habitación amplia en St. Antoni	Sant Antoni	69	1	155
2	Barcelona | Stunning Beach Apt....	Diagonal Mar i el Front Marítim ...	174	1	126
3	Terraza! Mediterráneo .10min ...	Sant Gervasi - Galvany	105	1	180
4	Nice and quiet room in the center	el Raval	66	1	298
5	Double Room with AC (Central ...	el Parc i la Llacuna del Poblenou	51	1	60
6	Single/ Room, Barcelona Excell...	Sant Pere, Santa Caterina i la Ri...	33	1	152
7	Twin Room Central + Close to t...	el Parc i la Llacuna del Poblenou	51	2	56
8	Comfort room Plaza España	Hostafrancs	46	2	210
9	Beautiful room with balcony	la Dreta de l'Eixample	100	2	143
10	Cozy room Plaza España	Hostafrancs	45	3	225
11	Cozy&Spacious Bedroom for Tr...	el Guinardó	44	3	76

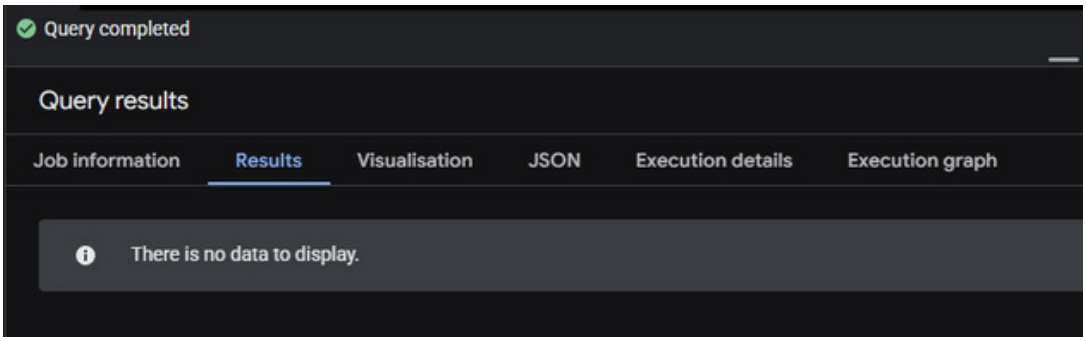
Insight: Highly booked but low-priced listings reveal clear opportunities for price optimization in Barcelona's Airbnb market.

6. Demand & Guest Behavior

High Price + High Demand (Goldilocks Zones)

```
WITH city_metrics AS (
  SELECT AVG(price) AS avg_price
  FROM `airbnb-sql-project.airbnb_barcelona.listings`
  WHERE price IS NOT NULL
)

SELECT
  neighbourhood_cleansed,
  ROUND(AVG(price), 2) AS avg_price,
  ROUND(AVG(availability_365), 2) AS avg_availability,
  COUNT(*) AS total_listings
FROM `airbnb-sql-project.airbnb_barcelona.listings`, city_metrics
WHERE price IS NOT NULL
  AND availability_365 IS NOT NULL
GROUP BY neighbourhood_cleansed
HAVING
  AVG(price) > avg_price
  AND AVG(availability_365) < 180
  AND COUNT(*) >= 100
ORDER BY avg_price DESC;
```



Insight: Barcelona does not have true “Goldilocks zones” where listings are both expensive AND consistently highly booked.

7. Guest Experience & Ratings

Ratings by Neighborhood

```
SELECT
 neighbourhood_cleansed,
 ROUND(AVG(review_scores_rating), 2) AS avg_rating,
 COUNT(*) AS total_listings
FROM `airbnb-sql-project.airbnb_barcelona.listings`
WHERE review_scores_rating IS NOT NULL
GROUP BY neighbourhood_cleansed
HAVING COUNT(*) >= 100
ORDER BY avg_rating DESC;
```



Row	neighbourhood_cleansed	avg_rating	total_listings
1	Diagonal Mar i el Front Marítim ...	4.7	127
2	la Vila Olímpica del Poblenou	4.68	116
3	el Guinardó	4.68	109
4	el Besòs i el Maresme	4.66	107
5	la Barceloneta	4.66	369
6	el Poblenou	4.66	373
7	el Parc i la Llacuna del Poblenou	4.65	158
8	la Dreta de l'Eixample	4.64	1830
9	Hostafrancs	4.64	165
10	el Camp d'en Grassot i Gràcia N...	4.63	238
11	les Corts	4.63	212
12	l'Antiga Esquerra de l'Eixample	4.61	767
13	la Sagrada Família	4.6	913
14	Sants	4.6	281

Insight: The relatively small variation in ratings suggests that Airbnb quality in Barcelona is consistently high, with location having only a limited impact on overall guest satisfaction.

Final Conclusion

Barcelona's Airbnb market is mature, competitive, and well-balanced, with clear segmentation across price, demand, and property types.

Key Takeaways:

- Entire homes dominate supply and revenue, but private rooms show strong demand efficiency
- Pricing is driven by capacity and competition, not just location
- The market exhibits a clear trade-off between price and demand, with no perfect "premium hotspot"
- Some listings are underpriced despite high demand, creating opportunities for revenue optimization
- The market is partially professionalized, with many hosts managing multiple listings
- Guest satisfaction is consistently high, with slight advantages in coastal and residential areas

Success in Barcelona's Airbnb market depends on balancing price, demand, and property positioning, rather than relying solely on location.

